

A Single Bout of Resistance Training Improves State Body Image in Male Weight-Trainers

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Although improvements in body image have been shown for longer-lasting resistance training, research on its short-term effects is currently lacking. The present study set out to test a hypothesized beneficial effect of a single bout of resistance training on global and specific (i.e., body fat- and muscularity-related) body image states of male exercisers. Additionally, a moderating effect of drive for muscularity was explored. In a controlled crossover study, 42 experienced weight trainers received (a) a session of resistance training; (b) a session of aerobic exercise (cycling); and (c) a session of magazine reading. Body image states were assessed before and immediately after each condition, and after 24 hr. As hypothesized, resistance training, but not cycling or reading, led to a significant increase in perceived muscularity on a silhouette measure of body image states ($g = 0.31$ or 3 kg). Both exercise conditions led to a significant decrease in perceived body fat ($g = -0.33$ or -2.5% for resistance training), but for resistance training, this only held true for men lower in drive for muscularity ($g = .58$, $p < .01$). After 24 hr, scores had returned to baseline levels. Even a single bout of resistance training yields short-term improvements in men's body image states on both the muscularity and fat dimension, which may act as a reinforcer of exercise behavior. The present findings might foster the understanding of body-image factors that contribute to the development of excessive weight-lifting behavior.

Keywords: exercise, resistance training, body image states, drive for muscularity, muscle dysmorphia

Resistance training (i.e., a mode of physical exercise designed to stimulate muscular fitness and/or muscle growth by working against ex-

ternal resistance) has become a popular indoor sport activity among boys and men in the U.S. and Europe (International Health, Racquet, & Sportsclub Association, 2015). Men's body image, that is, the perceptions, attitudes, and feelings surrounding their own and other men's bodies (Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999), may well have contributed to this trend, as large-scale surveys found that a substantial proportion of men are dissatisfied with their appearance (Frederick & Es-sayli, 2016), with respect to both body fat and muscularity (Jones & Crawford, 2005). The latter concern may be fueled by an increasing salience of an "ideal" male body (Thompson & Cafri, 2007) and, for some men, might result in a heightened drive for muscularity, that is, a trait-like desire to achieve a muscular and lean physique to feel better about

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one's body, of which resistance training is a behavioral manifestation (Edwards, Tod, & Molnar, 2014; McCreary, Hildebrandt, Heineberg, Boroughs, & Thompson, 2007).

Longer-lasting exercise programs have repeatedly been found to reduce body concerns and improve self-rated physical attractiveness both in women and men. For example, in their meta-analysis of 57 publications on body image effects of controlled aerobic and resistance training interventions, Campbell and Hausenblas (2009) calculated a small but robust positive effect from pre- to post- intervention, with larger effects for adult (as compared to school-aged) and female participants as well as for interventions with higher exercise frequencies. Moreover, another meta-analysis by Reel et al. (2007) suggested that resistance training may even yield larger positive changes in body image than aerobic exercise. At least in men, these changes seem to be based solely on subjective appraisals. Martin Ginis, Eng, Arbour, Hartman, and Phillips (2005) reported that for men participating in a 12-week strength-training program, an increase in self-rated muscularity, but not objective performance improvements, correlated with an increase in body satisfaction.

Although resistance training therefore seems to help women and men to feel better about their bodies in the long run, its potential short-term effects on men's body image states have never been studied (for affective responses see Greene & Petruzzello, 2015). This seems surprising for two interrelated reasons. First, the maintenance of exercise behavior is strongly influenced by short-term consequences such as enjoyment (e.g., Jekauc, 2015). Given the prevalent appearance concerns of boys and men (Frederick & Essayli, 2016), immediate positive changes to perceptual and evaluative/affective body image states via resistance training could constitute a potential maintaining factor. Second, short-term regulation of body image states by means of resistance training could act as perpetuating mechanism in muscle dysmorphia (MD), a clinical condition described in detail over the last few years (e.g., Pope et al., 2005). According to Pope, Gruber, Choi, Olivardia, and Phillips (1997), men suffering from MD are preoccupied with the idea of not being sufficiently muscular and "defined" (i.e., have an excessive drive for muscularity; McCreary & Sasse, 2000). This causes clinically significant

distress and a decline in role functioning because life becomes centered on a rigid exercise and diet regimen (see Lantz, Rhea, & Cornelius, 2002; for the related but not identical condition of bodybuilding dependence, see Smith & Hale, 2004).

To date, relatively little is known about how excessive resistance training in general, and in MD in particular, is developed and maintained (cf. Tod & Lavalley, 2010). Writing from an ethnographic perspective, Monaghan (2001) concluded from a series of interviews with bodybuilders that resistance training may entail immediate "representational and sensual pleasures that muscle enthusiasts derive from their vibrant physicality" (p. 331). On the one hand, experienced weight trainers may reframe muscle pain resulting from intense anaerobic exercise as indicating progress toward their goal of a fit-looking physique (Monaghan, 2001). On the other hand, somatosensory feedback of immediate exercise hyperemia (i.e., an increase in muscle blood flow) and muscle swelling (Korthuis, 2011) could also directly affect perceptual body image states (Cash, Fleming, Alindogan, Steadman, & Whitehead, 2002) insofar as weight trainers, including those suffering from MD, may perceive themselves to be more massive afterward, or, in bodybuilder lingo, to be "pumped up" (e.g., Gary, Butler, & Fiore, 1977). In line with this assumption, one study by Murray et al. (2012) found that patients with MD diagnoses, who did not differ from male anorexia patients with respect to rigid exercise practices and mood-related exercise withdrawal symptoms, did enjoy exercising significantly more than men with anorexia and even nonclinical controls.

Given the psychopathological similarities between muscle dysmorphia and anorexia nervosa (e.g., Murray, Rieger, Touyz, & De la Garza García, 2010), a model that provides a useful conceptual framework for the short-term regulation of body image states is the integrated cognitive-behavioral theory of eating disorders (Williamson, White, York-Crowe, & Stewart, 2004), which can be adapted to address muscularity concerns. The model assumes that an overconcern with body size or shape (e.g., with body fat or muscularity) acts as a psychological risk factor for the development of a dysfunctional, implicit body self-schema. Easily triggered by body- and self-related external and

internal stimuli (see Suchan, Vocks, & Waldorf, 2015), this body schema in turn biases the cognitive-attentional processing of these stimuli in a threatening fashion (i.e., amplifying preexisting fears of “being too fat” or “not being big enough”). The model predicts that women with weight concerns should especially show a biased processing of internal cues stemming from ingested food toward a “too fat” interpretation. By contrast, men with a pronounced drive for muscularity, who regularly work out with weights, may interpret a temporary absence of exercise-induced internal stimuli (e.g., hyperemia, stiffness, swelling) as experiential evidence (Williamson et al., 2004) that they are “not big enough.” To cope with the associated negative emotional state, such individuals may try to compensate for the lack of felt muscularity and the putative flaw in appearance (e.g., by resistance training). The resulting reduction of negative emotion may act not only to reinforce exercise behaviors, but also to further strengthen one’s negative body schema by ex post facto reasoning (e.g., “Weight lifting helped me feel better, therefore I am too small”). To sum up, it is hypothesized that the drive for muscularity is associated with a biased representation of one’s body, which can be temporarily altered by compensatory resistance training. It is further assumed that this process of body image state regulation establishes a vicious circle, in which the feeling of being “big enough” becomes tied to interoceptive stimuli from skeletal muscle tissue generated by resistance training.

Surprisingly enough, very few studies have addressed exercise-induced short-term changes to state body image in a controlled fashion. One of the first studies in this vein (Vocks, Hechler, Rohrig, & Legenbauer, 2009), conducted in female gym members, compared the immediate effects of a single bout of aerobic exercise on a bicycle ergometer on body image with a control condition (i.e., reading a newspaper) in a crossover design. As hypothesized, exercise led to a significant improvement ($d = 0.50$) in evaluative/affective body image state. Additionally, changes on a perceptive body image state measure (adjusting distorted photos of oneself) were positively correlated with trait drive for thinness, shape and weight concerns, and global body dissatisfaction ($r = .36-.52$; see also Scarpa, Nart, Gobbi, & Carraro, 2011). Perhaps even more relevant with regard to the present

topic, Thomas, Tod, and Lavallee (2011) used a controlled crossover design to psychometrically assess MD symptomatology in weight-lifters twice: once after a resistance training session and once on a rest day. In line with the adapted model presented above (Williamson et al., 2004), the authors found that participants scored significantly higher on drive for size, appearance intolerance, and functional impairment on a rest day than after resistance training ($d = 1.03, 0.69$, and 0.40 , respectively).

To sum up, there is preliminary evidence that state aspects of body image (Cash et al., 2002) in both women and men seem to be amenable to short-term improvements by exercise, a mechanism that has been proposed to act as a maintaining factor in eating disorders (Williamson et al., 2004). As yet, however, it is unknown whether this also holds true for muscularity-related perceptual body image states in men, whether this change is specific to resistance training, and whether men higher in drive for muscularity are especially prone to this effect.

The present study examined the immediate effects of a single resistance-training session on the body image states of males and compared them to an active and a passive control condition (cycling, reading). Following the model outlined above (Williamson et al., 2004), it was hypothesized that by providing the necessary internal cues, resistance training, but not aerobic exercise, would significantly improve perceived muscularity, but would not significantly improve the perceived proportion of body fat. It was further assumed that this perceptual body image effect would last to the following day, in parallel to delayed-onset muscle soreness, which peaks 24 to 72 hr postexercise (Cheung, Hume, & Maxwell, 2003). By contrast, following Vocks et al. (2009), a positive effect of cycling on perceived body fat was expected. Both modes of exercise were hypothesized to improve global evaluative/affective body image state (Vocks et al., 2009). Although there was no specific expectation for the reading control condition, previous findings by Boutcher and Landers (1988) and Vocks et al. (2009) suggest that reading would not improve body image states. Finally, in line with the model by Williamson et al. (2004), which assumes that body image concerns such as the drive for muscularity leave men at risk for a negative implicit body schema and a biased representation of internal

stimuli, and in accordance with results by Vocks et al. (2009) on the moderating role of drive for thinness on the effect of aerobic exercise in women, it was predicted that the effect of resistance training on perceived muscularity would be more pronounced in participants with a higher drive for muscularity.

Method

Participants

To address the hypothesis of an immediate positive body image reaction to resistance training and its moderation by drive for muscularity, a convenience sample of weight trainers was recruited. It was reasoned that this sample would be prone to such an effect and would show a heightened drive for muscularity (Edwards et al., 2014). Participants had to be aged 18 to 50 years and had to have worked out with weights at least twice a week over the preceding 6-month period (Day, McGuigan, Brice, & Foster, 2004). Exclusion criteria were a history of eating disorders, cardiovascular diseases, orthopedic disorders (e.g., spinal disk herniation), lingering athletic injuries, chronic conditions (e.g., diabetes mellitus), and acute infections. Potential participants were addressed via posters and leaflets placed in gyms, sporting equipment stores, and university buildings of a medium-sized university town in northern Germany. After making contact via e-mail, potential participants were called back and screened for inclusion and exclusion criteria. All participants gave written informed consent. Participants were blind to the hypotheses of the study and were told that the study aimed to examine objective exercise-related measures (e.g., heart rate) and their relation to well-being over different modes of exercise. They were also told that there would be an assessment of body composition by bioelectrical impedance. Upon completion of their individual assessments, all participants were thoroughly debriefed (including the possibility to withdraw their consent) and paid 12.50 € (about \$16.60 at the time of sampling) for their participation. If desired, they were able to receive a feedback protocol on objective measures (% body fat, heart rates during individual exercises).

To conduct a power analysis based on available empirical data, the intercorrelation ρ of the

dependent measures was estimated on the basis of the moderate to high test-retest reliability coefficients reported in previous studies (Cafri, Roehrig, & Thompson, 2004; Cash et al., 2002) as well as the correlation pattern and effect sizes from Vocks et al. (2009) (see Potvin, & Schutz, 2000; for the transformation of effect size η_p^2 provided by SPSS into f for G*Power, see Lakens, 2013). With $\rho = .55$, $f = .34$ and $1 - \beta = .80$, G*Power (Faul, Erdfelder, Lang, & Buchner, 2007), a required sample size of $N = 16$ was estimated for a minimum required two-group comparison (i.e., resistance training vs. reading, analogous to Vocks et al., 2009) with three repeated measures. To estimate the range of sample sizes potentially required, a conservative lower-bound model with $\rho = 0$ (i.e., uncorrelated dependent measures) and a medium effect size of $f = .25$ was calculated. Here, the estimated required sample size was $N = 54$.

Of $N = 54$ respondents screened, $n = 4$ did not meet the training criterion, $n = 2$ had to be excluded for health reasons, $n = 2$ were unable to find the time to participate, and $n = 4$ broke off contact after screening. Sampling was thus terminated at $n = 42$ (77.8%) due to the necessity for the sample size to be a multiple of six (see below).

Procedures

The study protocol was approved by the Institutional Review Board of the University's School of Human Sciences. The study applied a crossover design comprising three experimental conditions (see Figure 1): (a) resistance training on weight machines with standardized exercises, exercise sequence and repetition range at individualized intensities, (b) aerobic exercise on a bicycle ergometer with a standardized cadence range at an individualized intensity, and (c) reading a magazine from a standardized selection without sport- or body-related content. All conditions were separated by at least 1 day. In order to control for sequence effects, each participant received one of six possible sequences of the three conditions. Participants' assignment to one of these sequences was quasirandomized (i.e., followed a predetermined order) in order to balance the number of subjects assigned to each sequence. To rule out possible alternative explanations for the hypothesized body image state changes as far as possible, the

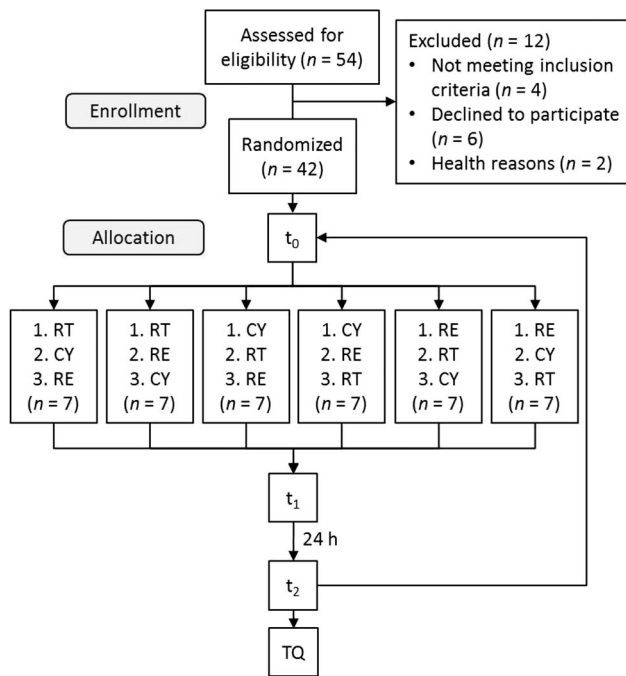


Figure 1. Participants' flowchart. t_0 = immediate pretest; t_1 = immediate posttest; t_2 = delayed posttest (+24 hr); TQ = Trait Questionnaires; RT = resistance training; CY = cycling; RE = reading.

opportunity for body checking in mirrors and social comparisons was minimized (Martin Ginis, Burke, & Gauvin, 2007). The gym of a judo training facility was selected for both exercise conditions because it was sparsely frequented, did not host a bodybuilding clientele, and its small number of mirrors could easily be kept out of the sight of exercising participants. The reading condition took place at a desk in a university lab room.

At the beginning and end of each of the three conditions, participants responded to state questionnaires three times (thus a total of nine times): immediately before (t_0) and after resistance training, cycling, or reading (t_1 , after normalization of monitored heart rate in exercise conditions) and 24 hr later (t_2) on their own after receiving a text message reminder via their cell phone. Before and after each day of testing, participants had to refrain from exercising for at least 24 hr to enable potential effects of previous exercise to be washed out and so that the t_2 assessment could be taken (i.e., a minimum of 48 exercise-free hours elapsed between conditions).

During t_0 – t_1 intervals, participants wore a heart rate monitor (Polar RS800CX), but were blind to their actual heart rate. After t_1 of the reading condition, body composition was assessed by bioelectrical impedance analysis (Maltron Body Fat Analyser BF 906), but participants remained blind to its result. The t_2 questionnaire battery of an individual participant's third (i.e., his last) condition also included the additional trait questionnaires described below. Debriefing, feedback, and payment took place on a separate day.

Four investigators (two female, two male), each with a bachelor's degree in psychology, conducted the testing. A certified fitness instructor was present at the facility and available at all times. Conditions were manualized, with the manual including verbatim instructions and pictures of exercises.

Resistance training. The design of the resistance-training intervention complied with evidence-based guidelines of the American College of Sports Medicine (ACSM; Garber et al., 2011; Ratamess et al., 2009) for programs targeting muscular hypertrophy in novice to inter-

mediate exercisers. For safety reasons, one-repetition maximum (1 RM) was not tested; thus, a range of eight to 12 possible repetitions was prescribed, roughly corresponding to a load of 70%–80% of 1 RM (Brzycki, 1993). According to the ACSM, this intensity of resistance training would be classified as submaximal but vigorous (Garber et al., 2011). Instructions specified that each set should be continued, without compromising form, to the point of failure or stopped only shortly before. To mitigate the potential for injury and reduce practice time, weight-training machines (gym80 International GmbH) were used for all exercises except crunches.

Each participant performed a fixed routine consisting of six exercises covering major muscle groups relevant to male body image (Ridgeway & Tylka, 2005). These were (a) bench presses on a Smith machine; (b) cable lat pull-downs; (c) horizontal leg presses; (d) cable tricep pushdowns; (e) seated machine bicep curls; and (f) crunches. After one or two practice and warm-up sets with 20 repetitions at a light intensity, resistance was increased, based on the participant's previous training experience, to a load that would limit the amount of repetitions to the prescribed range (i.e., eight to 12), allowing for an intermediate set if necessary. At this intensity, three working sets were performed, adjusting the load slightly if necessary. A rest interval of 3 min followed each working set to keep the number of repetitions within the prescribed range over multiple sets (de Salles et al., 2009). During sets, investigators kept out of the participants' sight, but participants were not explicitly led to believe that they were unobserved. Participants were allowed to drink water ad libitum between sets. After each resistance-training exercise, participants were asked to rate their perceived exertion (Day et al., 2004).

Aerobic exercise. As an active control condition, all participants exercised on a stationary bicycle (Ergo Cycle 1500; Ergo-Fit GmbH) for 45 min. Intensity (i.e., watts at cadences of 80–100 rpm) was tailored to individual cardiovascular fitness levels on the day of testing by means of an algorithm (OwnZone; Polar Electro) incorporating resting HR and HR variability (Achten & Jeukendrup, 2003), as assessed in a short pretest. The target HR zone was 65%–85% of an individual's maximum HR, which is considered a submaximal but vigorous intensity

(Garber et al., 2011). Participants were allowed to drink water ad libitum during cycling. Investigators kept out the participants' sight, but participants were not explicitly led to believe that they were unobserved. To match exercise conditions with respect to the frequency of social contact, investigators asked for an estimation of perceived exertion every nine minutes, for a total of five times.

Resting. As a passive control condition, participants read magazines for 45 min sitting at a table in a quiet office. The magazines comprised a standardized selection on geography, politics, cars, music, and animals, and did not include topics or pictures relating to fitness. They were allowed to drink water ad libitum. As a rating of perceived exertion was expected to seem odd to the participants, investigators checked in every 9 min to see how they were doing (e.g., whether they needed any more water), for a total of five times.

Measures

Subjective exercise characteristics. To control for subjective intensity of both exercise conditions, the Borg Category Ratio Scale version of the Rating of Perceived Exertion (CR-10 RPE) was used (Borg, Hassmén, & Lagerström, 1987) in the version validated by Day, McGuigan, Brice, and Foster (2004). The scale was translated into German by the study authors. The CR-10 RPE is an 11-point rating scale ranging from 0 (*rest*) to 10 (*maximal*). RPE scales have been successfully applied to resistance-training research several times in session and single-exercise versions, with ratings reflecting workloads (i.e., % 1 RM) and type of exercises (e.g., Sweet, Foster, McGuigan, & Brice, 2004). According to Day et al. (2004), the reliability of the CR-10 RPE is good, with an ICC of .88 for session scores from two different occasions with three different loads (50%–70%–90% 1RM) approximately 48 hr apart. In the present study, a sheet of paper (ISO 216 A4) with a vertically aligned RPE Scale including verbal anchors was presented to participants after each resistance-training exercise and during ergometer training to facilitate verbal ratings. Additionally, session-RPE scales were included in t_1 questionnaires of both exercise conditions.

An ad hoc item was used to assess similarity of exercise conditions to participants' habitual exercise regimen, ranging from 0 (*entirely different*) to 5 (*very similar*). The eight-item version of the Physical Activity Enjoyment Scale (PACES-8; Mullen et al., 2011), a 7-point semantic differential (e.g., *very pleasant* to *very unpleasant*), whose adjectives were translated into German for the present study, was used to control for exercise enjoyment. The authors reported a high internal consistency ($\alpha = .93$), which was only slightly lower in the present study ($\alpha = .85$ for aerobic exercise, $\alpha = .89$ for resistance training).

Global body image state. The Body Image States Scale (BISS; Cash et al., 2002) was administered to participants to assess the global evaluative and affective body image state (i.e., momentary satisfaction with one's size, shape, and weight, as well as feelings of physical attractiveness). The scale consists of six items with vertically aligned 9-point scales verbally anchored at each point (e.g., ranging from *extremely satisfied* to *extremely dissatisfied*). In the present study, items were coded such that higher scores indicate greater state body image dissatisfaction. Participants were instructed to report how they felt "at this very moment." For male participants, Cash et al. (2002) reported low to good internal consistencies ($\alpha = .62-.84$). For the German-language version administered in the present study, Cronbach's alpha was .85 in a sample of female gym-goers (Vocks et al., 2009). In the present sample, internal consistencies ranged from $\alpha = .82$ to .91.

Specific body image states. A modified version of the Somatomorphic Matrix (SM; Gruber, Pope, Borowiecki, & Cohane, 1999) was used to assess muscularity- and body-fat-related perceptual body image states. The original SM is a computerized two-dimensional contour-drawn silhouette scale, with 10×10 silhouettes of varying muscularity and body fat, from which respondents select their perceived and ideal body composition. As silhouettes were drawn according to photographs of males with known anthropometrics, selections can be translated into fat-free mass indices (Kouri, Pope, Katz, & Oliva, 1995) and % body fat (BF) values. Although SM actual-ideal discrepancy scores seem to be unreliable, test-retest reliability of the

self-muscularity score for a period of 7 to 10 days was acceptable ($r_{tt} = .78$; Cafri et al., 2004). The German-language version of the SM was successfully used with nonclinical males by Benninghoven, Tetsch, Kunzendorf, and Jantschek (2007), who found an actual-ideal discrepancy in fathers of female patients with eating disorder diagnoses. For the present study, the SM was applied in a paper-and-pencil version (Cafri & Thompson, 2004) that presented male silhouettes of reduced size (20×36 mm), arranged in a 10×10 matrix, on three fold-out sheets of paper (ISO 216 A3, 297×420 mm). To assess subjectively perceived muscularity and body fat, participants were first instructed to report how they felt "at this very moment" (Cash et al., 2002), followed by the usual SM instructions to select one's own actual and ideal physique. The latter acted as control variables for SM stability and its ability to discriminate between subjectively felt, wished-for and self-estimated "objective" body type.

Drive for muscularity. The Drive for Muscularity Scale (DMS; McCreary & Sasse, 2000; see Waldorf et al., 2014, for reliability and validity coefficients of the German-language version) is a 15-item trait measure of muscularity-related body image. Items are scored on a 6-point Likert scale from 1 (*always*) to 6 (*never*; scores are reversed so that higher scores indicate a higher drive for muscularity). For male samples, it is possible to calculate separate scores for a muscularity-oriented body image (MBI) subscale and a muscularity behavior subscale (McCreary, Sasse, Saucier, & Dorsch, 2004). For the present study, the former was selected as an operationalization of muscularity-specific attitudinal body image unfounded by exercise behaviors. DMS Item 10 on steroid use was not included. Cafri and Thompson (2004) reported high internal consistency and retest-reliability coefficients for both the total scale and subscales ($r_{tt} = .86-.89$; $\alpha = .84-.96$). In the present sample, the seven-item attitude subscale yielded an internal consistency of $\alpha = .88$.

Data Analysis

In a first step, sample and subjective exercise characteristics (RPE, similarity, and PACES

scores) were analyzed. Second, as an omnibus test of the effects of different forms of exercise on the dependent variables BISS, SM-perceived muscularity, and SM-perceived body fat, a $3 \times 3 \times 2$ doubly multivariate repeated measures ANOVA was conducted with the within-subject factors condition (resistance training, cycling, and reading) and time and the between-subjects factor muscularity-oriented body image (higher vs. lower, with DMS-MBI split at $Mdn = 3.07$). Accordingly, groups differed considerably on muscularity-oriented body image (lower DMS-MBI: $M = 2.28 \pm 0.59$; higher DMS-MBI: $M = 3.96 \pm 0.77$, $g = 2.41$). Third, this was followed by a series of $3 \times 3 \times 2$ univariate repeated measures ANOVAs, simple effects analyses, Bonferroni-adjusted pairwise comparisons and a planned contrast of Conditions (resistance training vs. the two control conditions) $\times$ Time (pretest vs. immediate posttest) to test whether resistance training is followed by an immediate effect on perceived muscularity that is superior to that of reading and cycling.

Greenhouse-Geisser correction was applied to degrees of freedom where Mauchly's test indicated a violation of the assumption of sphericity. Effect sizes were calculated as Hedges's g_s (between subjects) and g_{av} (within subjects; Lakens, 2013; conventionally, $d = 0.2, 0.5$, and 0.8 , $r = .10, .30$, and $.50$ and $\eta^2 = .01, .06$, and $.15$ are considered small, medium, and large effects, respectively; see also Fritz, Morris, & Richler, 2012, for their interpretation). For follow-up assessment of the reading condition only, data were incomplete for BISS (one case), and SM (two cases). For these cases, the first observation of the reading condition was carried forward.

Results

Sample

The mean age ($\pm SD$) of the 42 participants was $24.98 (\pm 6.57)$ years (range: 18–47), with 88.10% being under the age of 30. Twenty-four (57.14%) had completed high school, and 14 (33.33%) had completed college. Participants exercised $M = 3.60 (\pm 1.27)$ times per week for $M = 71.10 (\pm 23.12)$ minutes per day of exercise and had $M = 33.17 (\pm 1.27)$ months of training experience. Mean body mass index was $23.64 (\pm 1.99)$ kg/m², mean percentage of body

fat was $14.49 (\pm 3.90)$ %, and mean fat-free mass index was $20.16 (\pm 1.11)$. Mean resting heart rate before aerobic exercise was $69.24 (\pm 11.83)$ bpm. Drive for muscularity attitudes ($M = 3.12 \pm 1.08$) were uncorrelated with age and anthropometric or exercise characteristics. Table 1 presents the correlations of selected variables.

Subjective Exercise Characteristics

Mean session ratings of perceived exertion (RPE) were $4.21 (\pm 1.42)$ for resistance training and $3.90 (\pm 1.16)$ for cycling, corresponding to a “somewhat hard” estimation of both sessions. Session RPE ratings did not differ significantly across exercise conditions, $t(41) = 1.45$, $p = .16$. RPE scores for single resistance training exercises fell between 4.24 (tricep pushdowns) and 5.24 (bicep curls), with a mean of 4.52.

Mean enjoyment of exercise (PACES) was $5.12 (\pm 0.94)$ for resistance training and $4.58 (\pm 0.93)$ for cycling. The difference was statistically significant, $t(41) = 2.96$, $p < .01$, with a moderate effect ($g = 0.58$). Similarity with one's own exercise regimen was $2.48 (\pm 1.29)$ for resistance training and $1.93 (\pm 1.44)$ for cycling, $t(41) = 2.22$, $p < .05$, $d = 0.28$. To check for an association of subjective exercise characteristics with body image states, correlations of RPE, similarity ratings, and PACES scores with both BISS and the two SM depen-

Table 1
Correlations of Age, Anthropometric, Exercise, and Body Image Variables

	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1)	.19	.39*	.18	.35*	-.18	-.18	.31*	.11
(2)		.06	.23	-.21	-.07	.41**	-.04	-.23
(3)			.79**	.78**	-.26	.29	.67**	.41**
(4)				.29	-.23	.35*	.50**	.29
(5)					-.21	.06	.55**	.42**
(6)						-.16	.00	.25
(7)							.10	-.21
(8)								.61**

Note. (1) Age; (2) Training experience (month); (3) Body mass index; (4) Fat-free mass index (FFMI); (5) Body fat percentage (%BF); (6) Drive for Muscularity Scale, Muscularity-Oriented Body Image subscale; (7) Somatomorphic Matrix (SM), perceived FFMI (grand mean); (8) SM perceived %BF (grand mean); (9) Body Image States Scale (grand mean).

* $p < .05$. ** $p < .01$.

dent variables, including their t_0 – t_1 change scores, were calculated separately for the resistance training and the cycling condition. As was to be expected, 6% of coefficients became significant at the $p = .05$ level. None of these pertained to resistance training, body image difference scores, or RPE, and none survived controlling for family wise error (Bonferroni correction for correlations of an exercise characteristic with the dependent variables in each exercise condition: $.05/12$ [i.e., 3 dependent body image states $\times$ 3 assessment time points + 3 difference scores t_0 – t_1] = $.004$).

Effects of Exercise

MANOVA indicated significant effects of condition, $V = 0.37$, $F(6, 35) = 3.45$, $p < .01$,

$\eta_p^2 = .37$; time, $V = 0.50$, $F(6, 35) = 5.86$, $p < .001$, $\eta_p^2 = .50$; and Condition $\times$ Time, $V = 0.52$, $F(12, 29) = 2.61$, $p < .05$, $\eta_p^2 = .52$. The three-way interaction Condition $\times$ Time $\times$ Muscularity-Oriented Body Image (higher vs. lower) failed to reach statistical significance, $V = 0.37$, $F(12, 29) = 1.44$, $p = .21$, $\eta_p^2 = .37$. Table 2 provides descriptive statistics for the dependent variables.

Grand means were used to check for correlations among the three dependent body image states (Bonferroni correction: $.05/3$ grand M s = $.016$). Whereas BISS and SM-perceived body fat scores were correlated, $r = .61$, $p < .001$, both BISS and SM-perceived muscularity, $r = -.21$, $p = .18$ and the two SM variables, $r = .10$, $p = .54$ were uncorrelated.

Table 2
State Body Image Scores of Participants (*M*, *SD*, *Range*) in Three Experimental Conditions and Grand Means of Self-Reported Actual and Ideal Body Composition

Condition	Variable	Time			Differences	
		t ₀	t ₁	t ₂		
RT	BISS	3.65 (.95)	3.46 (1.02)	3.71 (1.03)	t ₀ > t ₁ ⁺ ; t ₂ > t ₁ [*]	
		2.00–5.33	1.33–5.33	1.83–5.83		
	SM FFMI-P	21.39 (2.57)	22.29 (3.08)	21.39 (2.39)	t ₀ < t ₁ ^{**} ; t ₁ > t ₂ ^{**}	
		18.00–27.00	18.00–28.50	18.00–27.00		
	SM %BF-P	17.24 (6.98)	14.76 (7.88)	16.76 (8.06)	t ₀ > t ₁ [*] ; t ₁ < t ₂ [*]	
CY	BISS	4.00–32.00	4.00–32.00	4.00–36.00	t ₀ > t ₁ ^{**} ; t ₁ < t ₂ [*]	
		3.73 (.90)	3.63 (.97)	3.74 (1.07)		
	2.17–5.67	2.17–5.83	1.83–6.17			
	SM FFMI-P	21.39 (2.77)	21.68 (2.82)	21.29 (2.72)		
		18.00–28.50	18.00–30.00	18.00–28.50		
SM %BF-P	16.57 (8.22)	14.29 (7.71)	17.14 (7.56)	t ₀ > t ₁ ^{**} ; t ₁ < t ₂ [*]		
RE	BISS	4.00–32.00	4.00–32.00	4.00–36.00	t ₀ < t ₁ ⁺ ; t ₀ < t ₃ ⁺	
		3.69 (1.02)	3.85 (.93)	3.95 (1.06)		
	1.83–6.33	2.33–6.33	1.83–6.17			
	SM FFMI-P	21.25 (2.50)	20.75 (2.34)	21.18 (2.48)		t ₀ > t ₁ [*]
		16.50–25.50	16.50–25.50	18.00–25.50		
SM %BF-P	17.43 (9.13)	17.33 (7.58)	17.71 (8.62)			
		4–40	8–32	4–36		
			BISS: RT < RE ^{**}			
			FFMI-P: RT > RE ^{**} ;			
			CY > RE ^{**}			
			%BF-P: RT > RE ^{**} ;			
			CY > RE ^{**}			
		Grand means				
	SM FFMI-E	SM FFMI-I	SM %BF-E	SM %BF-I		
	21.53 (2.41)	23.69 (2.77)	16.60 (7.45)	12.11 (5.56)		

Note. Values are given as $M(SD)$. t_0 = immediate pretest; t_1 = immediate posttest; t_2 = delayed posttest (+24 hr); RT = resistance training; CY = cycling; RE = reading; BISS = Body Image State Scale; SM = Somatomorphic Matrix (FFMI = Fat-free Mass Index; %BF = body fat percentage; P = perceived; E = estimated; I = ideal).
+ $p < .10$. * $p < .05$. ** $p < .01$.

Global Body Image State (BISS)

Follow-up ANOVA indicated significant effects of condition, $F(2, 80) = 3.52, p < .05$, $\eta_p^2 = .08$ and time, $F(2, 80) = 7.24, p < .005$, $\eta_p^2 = .15$. The Condition $\times$ Time interaction and the Condition $\times$ Time $\times$ Muscularity-Oriented Body Image interaction failed to reach statistical significance, $F(3.07, 122.88) = 1.87, p = .14$, $\eta_p^2 = .04$, and $F(3.07, 122.88) = 2.03, p = .11$, $\eta_p^2 = .05$, respectively. Post hoc simple effects analyses indicated significant differences between the three conditions only for scores from the immediate posttest (t_1), $F(2, 39) = 6.10, p < .01$, $\eta_p^2 = .24$ (with resistance training vs. reading: $g = 0.39, p < .01$). Significant changes within conditions over the three assessments were observed for resistance training, $F(2, 39) = 4.65, p < .05$, $\eta_p^2 = .19$, and reading, $F(2, 39) = 4.26, p < .05$, $\eta_p^2 = .18$, but not for the cycling condition, $F(2, 39) = 1.16, p = .32$, $\eta_p^2 = .06$. Inspection of profiles showed that for resistance training, evaluative/affective body image state improved nonsignificantly from t_0 to t_1 ($g = -0.19, p = .09$) before returning to the baseline level at follow-up ($p < .05$). By contrast, scores from the reading condition indicated a linear, nonsignificant deterioration in BISS scores from pretest to follow-up ($g = 0.25, p < .07$). To sum up, while cycling had no effect on men's momentary global evaluative/affective body image state, a significant but small postintervention difference between resistance training and reading was observed.

Perceived Muscularity (SM)

Follow-up ANOVA yielded significant effects of condition, $F(2, 80) = 7.26, p < .005$, $\eta_p^2 = .15$, and Condition $\times$ Time, $F(4, 160) = 5.16, p < .001$, $\eta_p^2 = .11$, on SM-perceived muscularity. The main effect of time failed to reach statistical significance, $F(2, 80) = 2.45, p = .09$, $\eta_p^2 = .06$, and the Condition $\times$ Time $\times$ Muscularity-Oriented Body Image interaction was also nonsignificant, $F(4, 160) = 1.39, p = .24$, $\eta_p^2 = .03$. According to post hoc simple effects analyses, conditions differed significantly in perceived muscularity immediately after the intervention, $F(2, 39) = 11.07, p < .001$, $\eta_p^2 = .36$ (with resistance training vs. reading: $g = 0.56, p < .01$). Pretest and follow-up scores were virtually identical. Scores changed signif-

icantly only in the resistance training and reading conditions, $F(2, 39) = 7.96, p < .005$, $\eta_p^2 = .29$, and $F(2, 39) = 4.78, p < .05$, $\eta_p^2 = .20$, respectively. Resistance training increased perceived fat-free mass from pretest to immediate posttest ($g = 0.31, p < .01$; cycling: $g = 0.10, ns$), whereas reading led to a decrease ($g = -0.21, p < .05$). As hypothesized, a significant contrast for the Condition (resistance training vs. control conditions) $\times$ Time (t_0 vs. t_1) interaction emerged, $F(1, 40) = 9.40 (p < .01)$, $\eta_p^2 = .19$. Finally, to test the discriminant validity of the SM perceived muscularity measure, the ANOVA was repeated twice with the SM ratings of estimated actual and ideal muscularity as dependent variables. Neither ANOVA yielded significant main or interaction effects.

To sum up, resistance training led to a significant, but short-lived, slight increase in perceived muscularity and was superior to the control conditions. The effect was specific to the aspect of perceived muscularity, as ideal muscularity and self-estimation of fat-free mass remained unchanged.

Perceived Body Fat (SM)

ANOVA indicated significant main effects of condition, $F(2, 80) = 4.45, p < .05$, $\eta_p^2 = .10$, and time, $F(2, 80) = 9.11, p < .001$, $\eta_p^2 = .19$. The Condition $\times$ Time interaction was not statistically significant, $F(2.93, 117.26) = 1.84, p = .14$, $\eta_p^2 = .04$, and the Condition $\times$ Time $\times$ Muscularity-Oriented Body Image interaction also failed to reach significance, albeit by a very small margin, $F(2.93, 117.26) = 2.55, p = .06$, $\eta_p^2 = .06$. Post hoc simple effects analyses yielded significant differences between conditions only immediately after the intervention, $F(2, 39) = 6.78, p < .01$, $\eta_p^2 = .26$, and significant changes only in the resistance training and cycling conditions, $F(2, 39) = 5.34, p < .01$, $\eta_p^2 = .21$, and $F(2, 39) = 6.17, p < .01$, $\eta_p^2 = .24$, respectively. The two exercise conditions did not differ significantly from each other in the immediate posttest ($g = 0.06, p > .90$), but participants felt significantly slimmer immediately after exercising than after reading in both the resistance training and the cycling condition, $g = 0.33, p < .01$, and $g = 0.39, p < .01$. Perceived body fat decreased from pretest to immediate posttest both in the resistance training and the cycling condition, $g = -0.33, p < .01$.

.05, and $g = -0.28$, $p < .01$, respectively, before returning to baseline levels at follow-up. It remained virtually unchanged in the reading condition. To sum up, exercise, irrespective of its mode, had a short-lived, slight positive effect on perceived body fat and was superior to a passive control condition.

To check for an undue influence of four outlying/extreme age cases (≥ 40 years), the follow-up ANOVAs were repeated after excluding these participants, which did not change the results in any meaningful way. Moreover, the analyses were rerun with investigator's sex as between-factor, which did not yield significant main or interaction effects.

Muscularity-Oriented Body Image (DMS) and Changes in Specific Body Image States

With respect to the size of the partial eta-squared of the three-way interaction in MANOVA ($\eta_p^2 = .37$) and the α level of the three-way interaction in ANOVA for SM-perceived body fat ($p = .06$), it was decided to also report results from simple effects analyses for men with higher versus lower muscularity-oriented body image scores, and to calculate effect sizes for t_0 - t_1 changes.

With respect to SM-perceived muscularity, both men with lower and higher DMS-MBI scores benefited from resistance training, with the higher-MBI group reaching statistical significance (lower MBI: $F(2, 39) = 2.76$, $p < .08$, $g = 0.28$; higher MBI: $F(2, 39) = 5.46$, $p < .01$, $g = 0.34$). Neither group significantly benefited from aerobic exercise (lower MBI: $F(2, 39) = 1.56$, $p = .22$, $g = 0.23$; higher MBI: $F(2, 39) = 1.23$, $p = .30$, $g = .06$).

With respect to SM-perceived body fat, lower-MBI exercisers benefited to a greater degree from resistance training than higher-MBI exercisers (lower MBI: $F(2, 39) = 6.34$, $p < .01$, $g = .58$; higher MBI: $F(2, 39) = 3.25$, $p = .05$, $g = .10$). In terms of difference scores, subjectively perceived body fat of lower-MBI men improved by -4% (± 5.44) on average (higher MBI: $M = -0.76 \pm 5.31$). The two groups were comparable in their reaction to aerobic exercise (lower MBI: $F(2, 39) = 3.00$, $p = .06$, $g = -.27$; higher MBI: $F(2, 39) = 3.25$, $p < .05$, $g = -.30$). To sum up, resistance training had a moderate positive effect on perceived body fat in men with a lower drive for muscularity.

Discussion

The present research is one of the first controlled studies to examine the effects of a single session of resistance training on the body image states of men, and is the first study to apply silhouette measures of perceptual body image states and an active exercise control condition. In doing so, it builds on and expands the ethnographic description of the immediate positive effects of bodybuilding by Monaghan (2001). It also extends the previous quantitative research on the short-term effects of exercise on body image by Vocks et al. (2009) on female gymgoers and by Thomas et al. (2011) on male weight trainers. In the present study, experienced lifters were asked to participate in single sessions of resistance training, cycling, and reading a magazine. Drawing on a cognitive-behavioral model of eating disorders that describes the regulation of negative affective states by compensatory behaviors such as exercise (Williamson et al., 2004), it was predicted that a single session of resistance training would yield an improvement in muscularity-oriented body image states of male weight trainers. In line with Vocks et al. (2009), a positive effect of both exercise conditions on momentary global evaluative/affective body image state was also expected.

For the latter hypothesis, support was weak and mixed. Although a significant difference between BISS scores immediately after resistance training versus reading was observed, improvements within exercise conditions were small and failed to reach statistical significance. The robust meta-analytic finding of a significant effect of longer-lasting exercise interventions on body image (Campbell & Hausenblas, 2009; Reel et al., 2007) thus does not seem to apply to single-session interventions with men and resistance training. However, the meta-analysis by Campbell and Hausenblas (2009) also found rather small body-image effects, especially for males and for normal-weight (vs. overweight) study participants. In fact, the small and non-significant effect of resistance training on evaluative/affective body image state observed in the present study is numerically identical to the effect for males from the aforementioned meta-analysis, which may point to a power problem for this analysis. Moreover, the present study applied the lowest possible exercise dose, and

Reel et al. (2007) found effect sizes to be significantly moderated by the total number of exercise sessions.

Results for the BISS were also markedly smaller in numerical terms than the BISS gain reported by Vocks et al. (2009), who also used bicycle ergometers, but targeted female gym-goers. This finding may simply mirror baseline differences in evaluative/affective body image between women and men on scales targeting women (e.g., Stanford & Lemberg, 2012). Moreover, the present sample was drawn from a population of nonobese male weight trainers, whose predominant reason for exercise does not seem to be general physical appearance (as assessed by four of six items of the BISS), but rather developing their bodies from “scrawny to brawny” (cf. Mejia & Berardi, 2005). Thus, the BISS may not be as valid for assessing resistance-training-induced changes in evaluative/affective body image state of males as it is for women performing aerobic exercise.

Given recent conceptualizations of male body image (e.g., Jones & Crawford, 2005) as encompassing the dimensions of body fat and muscularity, a silhouette measure was applied in order to assess state-like aspects of these dimensions. Supporting the main hypothesis, only resistance training led to a significant increase in perceived muscularity, but both modes of exercise fostered a feeling of leanness (at least for men lower in drive for muscularity attitudes). On average, silhouettes selected immediately before and after resistance training gained about 3 kg of muscle mass and lost 2.5% body fat.

For the first time, this finding lends credence to the notion of a “pumped-up,” subjectively more muscular body after exercise, a state that may act as a potential short-term reinforcer of resistance training. Although the present results stem from a non-MD group, they seem to fit with the model by Williamson et al. (2004), which sees exercise as a compensatory behavior, acting to counter inner negative cognitive-affective states (e.g., feeling “not muscular enough”). By providing preliminary evidence for this mechanism of short-term body-image regulation—albeit with nonclinical men—the present research is in line with recent claims that MD concepts should be informed by learning theory and that the condition is not sufficiently understood by only looking at symptoms

and distal etiological factors (Tod & Lavallee, 2010). It is therefore tempting, although still very speculative, to assume that operant conditioning contributes to the development and maintenance of excessive, rigid, and dependence-like resistance training habits (e.g., Smith & Hale, 2004). Although Williamson et al.’s (2004) model suggests that this mechanism consists of the temporary alleviation of negative affect (i.e., negative reinforcement; see also Thomas et al., 2011), it cannot be ruled out that the observed increase in subjective muscularity might also act as a direct positive reinforcer of resistance-training behavior, contributing to its maintenance over the longer term. This interpretation fits with the finding that male patients with MD diagnoses reported enjoying exercising significantly more than those with anorexia nervosa (Murray et al., 2012). For future studies addressing the contribution of short-term body image reinforcers to the establishment of excessive exercise behaviors, it would therefore be a logical next step to follow a cohort of novice weight trainers to test for longitudinal associations of early body-image state responses with exercise behaviors (i.e., frequency, adherence) over time. At the same time, the influence of intrinsic motivation (e.g., enjoyment, social contacts) on resistance training behavior should be controlled for, as previous studies have found intrinsic motives, but not extrinsic, appearance-oriented motives, to be associated with exercise behavior (Jekauc, 2015; Ryan, Frederick, Lepes, Rubio, & Sheldon, 1997).

With respect to perceived body fat, both exercise conditions exerted a significant positive effect. Contrary to the hypothesis, there was no evidence suggesting an inferiority of resistance training in modifying fat-related body image state. There are several possible explanations for this finding. Interoceptive changes due to resistance training (e.g., increase in blood flow, swelling of the muscles, soreness) could directly give the impression of being “more cut up” and less “flabby” (Ridgeway & Tylka, 2005, p. 213) or could divert attention away from body fat (i.e., induce an attentional bias toward muscularity), and may thus change perceived body composition without necessarily changing perceived muscle mass.

This unexpected change in perceptual body image state may act as an additional, yet overlooked, reinforcer of resistance training, which

also corresponds with a specific reason for exercising. Some men specifically wish to get “ripped and cut,” that is, want their muscles to show through but do not necessarily desire to add muscle mass. It could be argued that in samples of nonobese men especially (in the present study, 90% of participants had body fat percentages ≤ 20), it is this “drive for leanness,” rather than a more general drive for thinness (Smolak & Murnen, 2008), that may be tapped into by the body fat dimension of male-specific silhouette measures displaying increasingly visible abdominal muscles and striation, and that may be temporarily satisfied by exercise.

The secondary hypothesis of a moderating role of muscularity-oriented body image for resistance-training-induced changes in perceived muscularity could not be confirmed, that is, men with a higher and a lower drive for muscularity benefitted to a similar degree. One possible explanation for this finding is that drive for muscularity, in contrast to what the model by Williamson et al. (2004) postulates for weight concerns, is not (or at least not in this nonclinical sample) associated with an implicit body self-schema which biases the representation of internal cues and necessitates “compensatory” exercise behavior. Although previous studies found that drive for muscularity was correlated with measures of body image problems such as social physique anxiety (e.g., Brunet, Sabiston, Dorsch, & McCreary, 2010), in the present sample, it was uncorrelated with perceptions of actual muscularity or body fat. In line with this, in an eye-tracking study in a nonclinical sample of weight trainers, Cordes, Vocks, Düsing, Bauer, and Waldorf (2016) found that whereas drive for thinness was indeed associated with an attentional bias toward one’s “problem areas,” drive for muscularity was associated with a bias toward one’s positively valenced body areas. It thus seems that drive for muscularity, although being a core feature of MD, does not per se constitute a psychological risk factor sensu Williamson et al. (2004). Accordingly, future studies looking at short-term effects of resistance training on perceptual body image could, in a first step, include measures of MD symptoms such as exercise dependence and social physique anxiety (Rhea, Lantz, & Cornelius, 2004) in order to identify weight trainers with subclinical symptomatology, a strategy employed in eating disorders research (e.g., Jansen, Nederk-

oorn, & Mulken, 2005). In a second step, the reactions of patients with MD diagnoses to single bouts of resistance training could be compared to those of nonclinical weight-trainer controls.

Surprisingly, a change in body fat related perceptual body image state was restricted to participants with a lower drive for muscularity, who improved with a moderate effect, which only failed to reach statistical significance by a very small margin. It is possible that during and after resistance training, exercisers with a higher drive for muscularity are interoceptively focused on, or differentially interpret, bodily stimuli congruently with their expectation of muscle gains. Weight trainers with a lower drive for muscularity, on the other hand, exhibited more general body image improvements, in line with a less muscularity-focused approach to resistance training.

And finally, the hypothesis that an effect of resistance training on perceived muscularity would last at least 24 hr was not supported at all. It thus seems that, once exercise hyperemia and other immediate adaptive physiological responses have fully abated after a couple of hours (Korthuis, 2011), delayed-onset muscle soreness alone is not sufficient to produce longer-lasting body image state improvements. Alternatively, on a rest day, evaluative body image factors may trump perceptual aspects, with weight trainers feeling less muscular just because they know they have not exercised.

The present study has several methodological limitations worth mentioning. First, due to its proof-of-concept character, it targeted rather experienced weight trainers from a nonclinical population, which limits the ability to generalize the findings to novice exercisers and to men with MD diagnoses. Second, with two active exercise conditions applied in a crossover design, differential carryover effects, although unlikely, cannot be ruled out, a problem that could have been countered by longer washout periods (Kantowitz, Roediger, & Elmes, 2015). Third, as it does not comprise muscularity items, the BISS as a measure of evaluative/affective body image state might not be as useful for studies on resistance training in males as it is for research on aerobic exercise conducted for weight-related reasons, or for studies on body image in women. For the purpose of the present study, scales addressing momentary and specific

thoughts and feelings on one's muscularity or scales adapted to assess state-like aspects of MD symptomatology (Thomas et al., 2011) might have been better suited. As no muscularity-oriented measure of evaluative/affective body image state has been established to date, the present study applied the BISS as a validated and widely used state measure (Cash et al., 2002). Fourth, although adequately powered for the detection of effects of resistance training on perceived muscularity, the present study seems to have been slightly underpowered for the detection of the empirically observed small effect on evaluative/affective body image.

Fifth and finally, for this pilot study, care was taken to protect internal validity by standardizing exercise regimens and ruling out alternative explanations for possible effects of resistance training. Inevitably, this prescription-based approach (Ekkekakis, 2009) reduces the ecological validity and probably autonomous motivation and self-efficacy of participants (e.g., Slovinec D'Angelo, Pelletier, Reid, & Huta, 2014). In resistance training, there is a host of variables (e.g., volume and intensity, full-body vs. split routine, free weight vs. machines, and inclusion of intensification techniques) that allow experienced exercisers to individualize their workout routine. Possibly as a result of standardization, participants reported only a moderate similarity with their own typical exercise protocol, and both session and single-exercise ratings of perceived exertion were lower than published scores for a comparable exercise protocol and intensity (Sweet et al., 2004). It is therefore possible that individualized routines with self-selected intensities (Ekkekakis, 2009) might have produced greater similarity and RPE scores and more pronounced improvements in body image states.

To conclude, the present study is the first to provide evidence for a short-lived effect of a single resistance training session on the body image states of male weight trainers beyond that of aerobic exercise. It can be speculated that this "greatest feeling you can get in the gym" (see Gary et al., 1977) contributes to the short-term reinforcement of resistance training and possibly the establishment and maintenance of excessive exercise patterns in MD. If future studies substantiate this notion, a deeper understanding of this mechanism could aid exercise professionals in tailoring interventions to maximize body satisfac-

tion after resistance training (Martin Ginis et al., 2005) and could inform the development of treatment options (e.g., alternative reinforcement; training of emotion regulation strategies) and case formulations for MD and related body image problems.

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